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Review

Regenerative Braking Systems in Electric Vehicles: A Comprehensive Review of Design, Control Strategies, and Efficiency Challenges

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Abstract: Regenerative braking systems (RBS enhance energy efficiency and range in electric vehicles (EVs) by recovering kinetic energy during braking for storage in batteries or alternative systems. This literature review examines RBS advancements from 2005 to 2024, focusing on system design, control strategies, energy storage technologies, and the impact of external and kinematic factors on recovery efficiency. Based on a systematic analysis of 89 peer-reviewed articles from Scopus, it highlights a shift from basic PID controllers to advanced predictive algorithms like Model Predictive Control (MPC) and machine learning approaches. Technologies such as brake-by-wire and in-wheel motors improve safety and stability, with the latter excelling in all-wheel-drive setups over single-axle configurations. Hybrid Energy Storage Systems (HESS), combining batteries with supercapacitors or kinetic accumulators, address power peak demands, though cost and complexity limit scalability. Challenges include high computational requirements, component reliability in harsh conditions, and lack of standardized testing. Research gaps involve long-term degradation, autonomous vehicle integration, and driver behavior effects. Future work should explore cost-effective HESS, robust predictive controls for autonomous EVs, and standardized frameworks to enhance RBS performance and support sustainable transportation.

Keywords: Regenerative Braking System; energy recovery; electric vehicle; energy efficiency; regenerative braking

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1. Introduction

Electric vehicles (EVs) play a pivotal role in the transformation of the transportation sector towards sustainability, offering the potential to reduce greenhouse gas emissions and reliance on fossil fuels. A key factor in enhancing the energy efficiency and range of EVs is the energy recuperation system, which allows for the recovery of kinetic energy during braking and its conversion into electrical energy stored in batteries. While this technology has been utilized in hybrid and electric vehicles for years, it continues to face challenges related to optimizing energy recovery in varying operating conditions, integrating with mechanical braking systems, and overcoming limitations imposed by battery and electric motor properties. With growing scientific and industrial interest, the last two decades have witnessed a dynamic development in research on energy recuperation systems, which is reflected in the exponential growth of publications in this area.

This review paper aims to provide a synthetic analysis of the scientific literature concerning energy recovery systems in electric vehicles, with a particular emphasis on their structure, control strategies, energy storage technologies, and the influence of external factors and kinematic parameters on recuperation efficiency. Based on a systematic review

of publications from 2005–2024, key trends have been identified, such as the development of predictive algorithms and hybrid energy storage systems, as well as challenges, including technical limitations and the lack of uniform design standards. This review distinguishes itself through an interdisciplinary approach, integrating technological, computational, and mechanical aspects, which enables a comprehensive assessment of the state of knowledge. Its objective is not only to summarize existing achievements but also to identify research gaps and propose directions for future research. In this way, the paper makes a significant contribution to the development of EV technology, supporting the design of more efficient and safer recuperation systems, which is important for both science and the automotive industry.

2. Fundamentals of Energy Recuperation Systems: Design and Operation

In vehicles with an electric drive system, it is possible to recover a portion of the kinetic energy during braking. From the perspective of electric vehicles, this is currently the only opportunity to replenish energy while driving. Both conventional vehicles and those with electric drive systems must meet a series of homologation requirements to be approved for road use, including specifically detailed regulations related to vehicle braking.

The structural diagram of a regenerative braking system in electric vehicles is shown in Figure 1. Regenerative braking involves the conversion of kinetic energy into electrical energy during vehicle deceleration or braking. When the braking system is applied, the electric machine operates in generator mode, reversing the direction of torque and energy flow, thereby transforming mechanical energy into electrical energy (indicated by the green line). The energy transferred to the battery is regulated by the Battery Management System (BMS). The intensity of regenerative braking is influenced by numerous factors that can limit the amount of recovered energy. These include battery-related parameters such as the State of Charge (SOC), ambient temperature, and charging rate. Additionally, the power, speed, and torque of the electric motor, along with its efficiency, play a crucial role. The regenerative braking algorithm operates based on signals from measurement sensors (brake pedal position, vehicle velocity), and the braking intensity is determined by real-time calculations performed by the Electronic Control Unit (ECU). The system also includes a mechanical brake subsystem (indicated by the red line) that ensures adequate braking performance when energy recovery is limited.

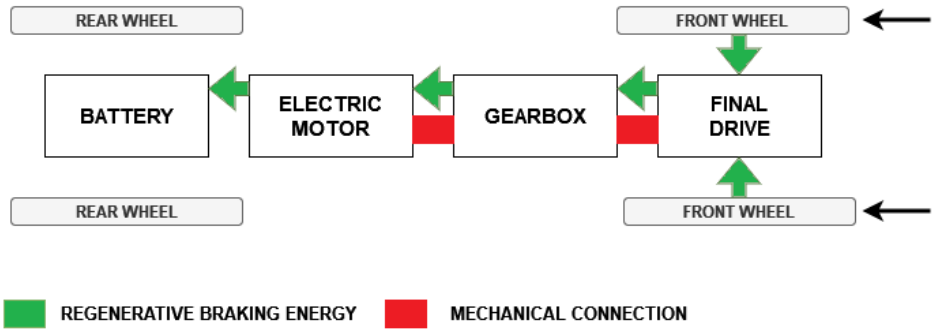


Figure 1. Layout of regenerative braking system in a front-wheel drive electric vehicle.

The energy recovery system in electric vehicles comprises the following components:

- an electric machine that operates in generator mode during braking, converting the vehicle's kinetic energy into electrical energy;
- an energy storage that stores the energy recovered during braking. Typically, this is an electrochemical battery, but solutions involving supercapacitors, kinetic energy storage systems, or hydropneumatics accumulators as supplementary energy storage devices are also encountered;

- a controller, which manages the energy recuperation process, distributing energy between the regenerative system and mechanical brakes to ensure safety and braking efficiency;
- sensors that monitor system operating parameters, such as vehicle velocity, brake and accelerator pedal positions, electric motor speed, and battery state of charge;
- software that controls the controller's operation based on sensor data and other parameters.

At low rotational speeds, the maximum torque is within a constant range. However, a decrease in rotational speed causes the braking torque available for energy recovery to diminish to zero. If the generator reaches its torque limit, as the rotational speed decreases, there is a continuous reduction in the friction torque of the braking system due to the increase in the generator's torque. In the final stage of braking, the generated torque fades and is replaced by friction torque. At low rotational speeds, the electric generator is unable to produce generated torque. The control system continuously monitors all critical system parameters, determining the precise moment when energy recovery during braking occurs.

During the coordination and calculation of regenerative braking participation, two primary limitations are taken into account. The first limitation is the maximum regenerative braking capacity, which is typically determined by the braking torque capability of the electric machine in generator mode. The second limitation is the low efficiency of the regenerative braking process at low velocity. This results from insufficient voltage generated by the electric machine [1].

Due to the limited power of the energy recovery braking system, a mechanical braking system is still required. The interaction between the regenerative braking system and friction brakes in electric vehicles introduces complexity in optimizing the operation of the regenerative braking system. One of the main challenges is managing the distribution of braking force between the regenerative and mechanical braking systems to achieve the highest possible energy recovery from regenerative braking. Another challenge is the method of regulating the distribution of braking force to the front and rear wheels to ensure effectiveness and safety during braking. Consequently, the control strategy for mechanical brake systems and the energy recuperation system significantly influences the effectiveness of the braking system in electric and hybrid vehicles. Appropriate control strategy algorithms are designed to find a combination of regenerative and friction braking that maximizes the amount of recovered energy while providing the necessary braking force demanded by the driver, maintaining vehicle stability and user safety. The regenerative braking algorithm operates based on signals from measurement sensors, providing information on parameters such as the position and angular position of the accelerator and brake pedals, vehicle velocity, and battery energy level [2,3]. The efficiency of regenerative braking is influenced by the result of real-time calculations.

Two primary regenerative braking systems are distinguished in electric vehicles: parallel and series (Figure 2). These systems vary in control complexity.

In Figure 2, the two regenerative-braking architectures are illustrated in detail. In the parallel system (Figure 2a), the electric motor (acting as a generator) provides up to its rated braking torque based solely on driver command and vehicle speed, while the mechanical friction brakes remain inactive until the generator torque limit is reached. Once the regenerative torque setpoint exceeds this threshold, the Electronic Control Unit (ECU) commands the hydraulic actuation of the friction brakes. In contrast, the series system (Figure 2b) fully integrates the regenerative generator, hydraulic friction brakes, and the ABS module under ECU supervision: at low deceleration demands only the motor is engaged; as driver-requested torque increases, the ECU issues CAN-bus signals to both the motor inverter and ABS valve unit, ensuring coordinated torque blending and wheel-slip control [4–6].

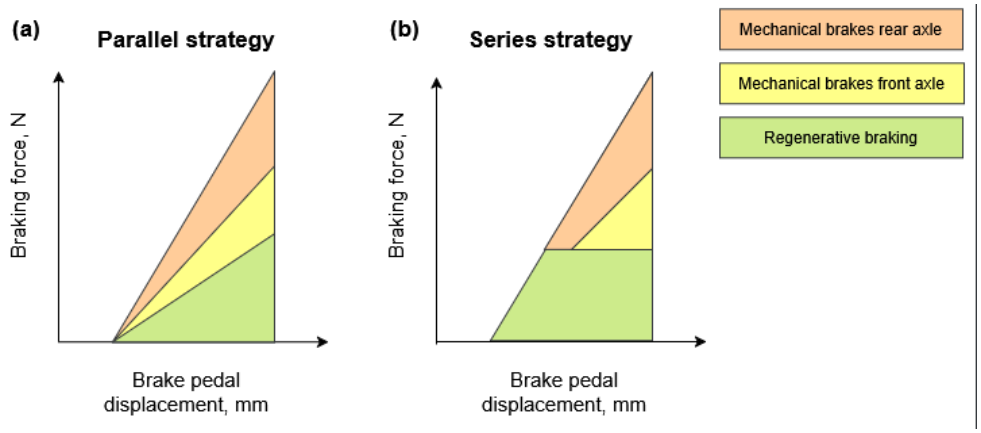


Figure 2. Braking force distribution strategies between mechanical brakes and energy recovery system, (**a**) parallel, (**b**) series.

Regenerative braking in both electric and hybrid vehicles is the subject of much research and analysis. Numerous works present original models of energy distribution control strategies during braking. Analyzing the literature on energy recovery systems in electric vehicles, several main issues can be distinguished, which are addressed by researchers. The review indicates that the most frequently addressed topics include: algorithms in energy recovery system control (electric motor control, controller operation control), integration of the energy recovery system with the friction braking system, recovered energy storage systems, electric machines, and materials used in friction brakes.

3. Systematic Literature Review Methodology

In conducting the literature review on regenerative braking systems in vehicles, the Scopus database was utilized. Initially, the search scope was defined by entering the phrase "regenerative braking system" into the search options. In the Scopus database, 482 scientific articles were identified. Further searches were conducted using the keywords "energy recovery" and "electric vehicle", with the search narrowed to works exclusively in the English language. TITLE-ABS-KEY ((regenerative AND braking AND system) AND (energy AND recovery) AND (electric AND vehicle)) AND (LIMIT-TO (LANGUAGE, "English")). Ultimately, 474 papers were retrieved. In the subsequent stage, the abstracts were reviewed, and research papers pertaining to passenger vehicles were selected. Finally, 89 scientific articles were utilized for the review, which were then meticulously examined (Figure 3). Each full article was read to further assess whether it provided significant insights relevant to this review.

The aim of this review was to comprehensively analyze and synthesize the available scientific literature concerning energy recovery systems in vehicles with electric drive systems. The key assumptions are:

- chronological compilation of research development—identifying trends in research and technology implementation of energy recovery, based on publications from 2005–2024;
- defining research areas—characterizing interdisciplinary approaches, encompassing various areas and scientific fields;
- diversification of vehicle types for which analyses were conducted;
- identification of research tools and analytical, modeling, and simulation methods used in research;
- methods for optimizing the regenerative braking process to increase the level of recovered energy;

- solutions aimed at improving the safety and stability of vehicles with energy regeneration systems during braking;
- evaluation of recuperated energy storage technologies: applied solutions, including hybrid systems, and their impact on enhancing the energy recovery system's efficiency;
- determination of the influence of external factors and kinematic parameters on the efficiency of energy recovery systems;
- providing conclusions and recommendations, suggesting technological improvements, and pointing out potential future research areas related to energy recovery systems.

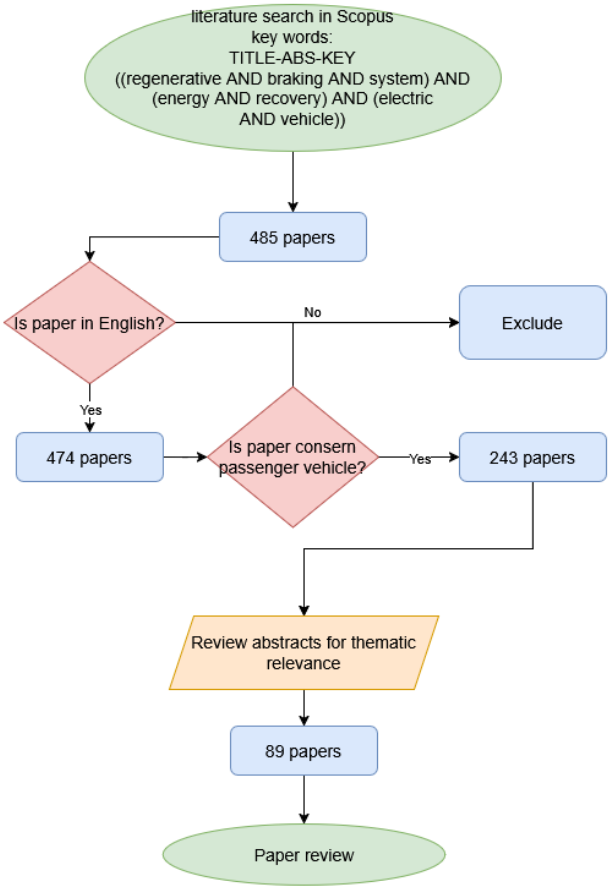


Figure 3. Literature review methodology process.

This review aims to identify current achievements, key challenges, and research gaps that can serve as a starting point for future innovations in energy recovery technologies for electric vehicles.

4. Bibliometric Analysis

The number of publications on regenerative braking in electric vehicles shows a clear upward trend over the years. Until 2000, only 7 papers were published, indicating an early stage of research on this technology (Figure 4). In the years 2000–2005, the number of publications increased slightly to 8, and then more noticeably in the years 2006–2010, reaching 39 papers, which was associated with the development of hybrid and electric drive systems in vehicles. A dynamic increase occurred in the years 2011–2015, when the number of publications rose to 96, and the most significant development occurred in the years 2016–2020, when 162 papers were published, reflecting the rapid development of technology and the growing interest in its optimization. After 2020, the number of papers remained at a high level of 161, indicating a stable interest among researchers in this topic and the continued development of energy recuperation systems for electric vehicles.

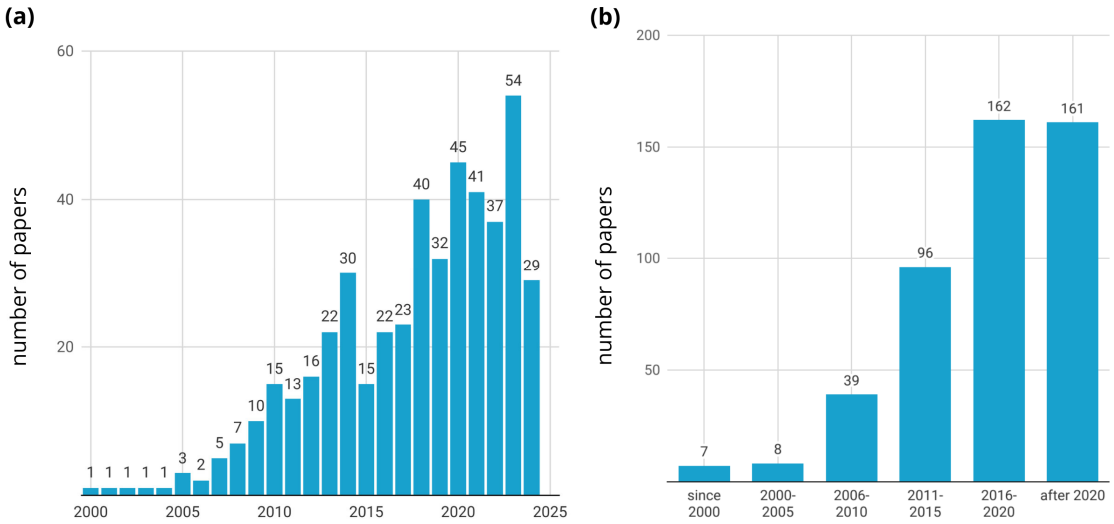


Figure 4. Number of papers on energy recuperation systems in electric drive vehicles: (**a**) annual and (**b**) five-year trends.

Below are the 5 most frequently cited papers, indicating their significant impact on the field:

1. "Modelling and control of hybrid electric vehicles" DOI: 10.1016/j.rser.2017.01.075 (281 citations, 2017)—a publication on hybrid vehicle systems and energy control strategies;
2. "Energy management strategy based on fuzzy logic for hybrid electric vehicles" DOI: 10.1016/j.jpowsour.2008.06.083 (234 citations, 2008)—this paper presents a fuzzy logic-based approach to optimizing energy transfer in hybrid vehicles;
3. "Development of brake system and regenerative braking for EVs" DOI: 10.1109/TVT.2014.2325056 (176 citations, 2015)—paper focuses on the integration of mechanical brake systems and energy recovery systems in electric vehicles;
4. "Model predictive control-based efficient energy recovery" DOI: 10.1016/j.enconman.2015.12.077 (174 citations, 2016)—study presents predictive control models used in energy recovery systems in hybrid vehicles;
5. "Investigation of regenerative braking systems for hybrid electric vehicles" DOI: 10.4271/1999–01-2910 (162 citations, 1999)—one of the earliest studies on regenerative braking in hybrid vehicles.

The analysis of keywords from the reviewed papers allows for the identification of the main thematic areas addressed in the context of energy recuperation systems in electric vehicles. Based on the visualization of keywords using the VOSviewer tool, the most frequently occurring issues related to regenerative braking systems in electric vehicles within the analyzed Scopus database papers were examined (Figure 5). The graph illustrates the network of connections between terms, enabling the identification of key research areas and potential development directions for this technology.

The VOSviewer map (Figure 5) visually represents keyword analysis: node size corresponds to keyword frequency, and the thickness of connecting lines indicates the strength of their co-occurrence. Four primary clusters are evident: (1) Control Strategies (red), prominently featuring "model predictive control", "fuzzy control", and "control strategy", signifying substantial research into advanced algorithms. (2) Energy Storage (green), highlighting "supercapacitor", "ultracapacitor", and "energy storage", which underscores a strong focus on high-power buffering solutions. (3) Vehicle Platform & Simulation (blue), centered around "electric vehicle", "simulation", and "energy efficiency", illustrating the widespread application of modeling tools in the field. Lastly, (4) Mechanical

& Stability (purple), encompassing "braking stability" and "braking force distribution", points to efforts in seamlessly integrating regenerative and friction braking for enhanced safety. Notably, the proximity and strong linkages between the control-strategy and energy-storage clusters suggest emerging hybrid approaches. Furthermore, connections to "hybrid electric vehicle" and "fuel cell" nodes hint at valuable cross-technological insights.

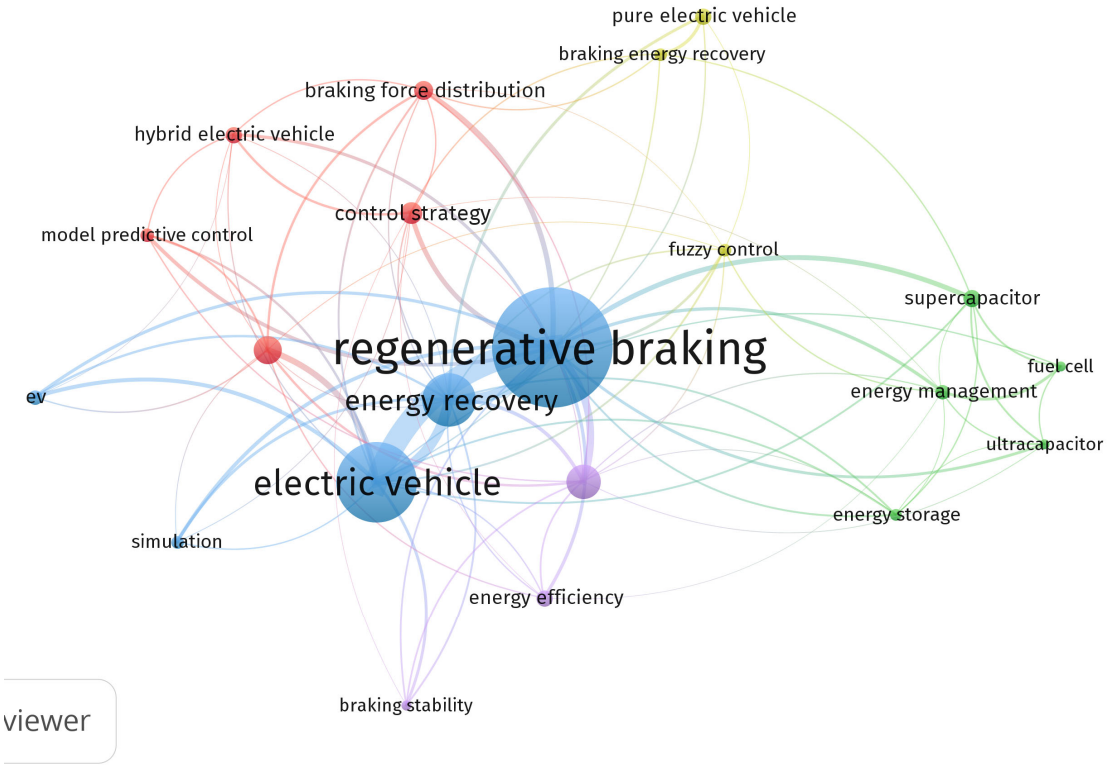


Figure 5. Keyword co-occurrence in electric vehicle energy recovery studies.

The Scopus database search was conducted based on 3 keywords: "regenerative braking system", "energy recovery", and "electric vehicle". The analyzed publications focus on research into the efficiency of the energy recovery process during braking and the integration of this technology with energy management systems. Issues related to the distribution of braking forces and vehicle stability during regenerative braking are discussed. Many works address the optimization of regenerative braking systems using advanced algorithms, such as Model Predictive Control and Fuzzy Control. Numerous publications also focus on recovered energy storage systems, such as supercapacitors. The use of simulation tools to analyze and optimize the performance of regenerative braking systems under various operating conditions is prevalent in the literature. In the analyzed thematic area, many works present research not only for electric vehicles. Numerous studies address regenerative braking and recovered energy for hybrid vehicles and fuel cell vehicles.

In summary, the most frequently addressed research topics indicate an interdisciplinary approach to regenerative braking, encompassing technological (energy storage), computational (control strategies), and mechanical (braking stability) aspects. These studies play a crucial role in the further development of electric vehicles, contributing to their efficiency and safety.

Recent publications (from 2020–2024) continue to focus significantly on:

- regenerative braking systems—the primary emphasis remains on how braking systems in electric and hybrid vehicles can efficiently recover energy. This is evidenced by the frequent use of words such as "braking" (159), "regenerative" (110), and "energy" (95);

- control strategies—ongoing research also addresses advanced energy management control systems, as indicated by frequent mentions of the words “control” (84) and “strategy” (64);
- optimization and simulation—recent studies increasingly emphasize the optimization of vehicle systems and the use of simulations to improve vehicle design.

5. Key Research Areas in Regenerative Braking Technology

5.1. Optimization Methods for Enhancing Regenerative Braking Efficiency

The efficiency of energy recovery in regenerative braking systems in electric and hybrid vehicles depends on control strategies that balance the proportions between regenerative and mechanical braking. The goal of these strategies is to maximize kinetic energy recovery while ensuring vehicle safety and stability under varying operating conditions. Much of the reviewed research is dedicated to the identification of algorithms for electronic control systems that maximize the efficiency of regenerative braking in electric vehicles. Electronic control systems govern this process using specific parameters, and several control algorithm types are utilized.

The optimization of regenerative braking systems relies on the dynamic distribution of braking force between the electric motor, operating as a generator, and the mechanical friction brakes. The literature indicates several key control parameters, such as vehicle velocity, motor torque, battery state of charge, and brake system pressure.

A basic strategy utilizes vehicle velocity as an operational variable [7–9]. Regenerative braking is activated above a certain threshold, while at low velocity (<10 km/h), only mechanical brakes are employed. [10]. Research indicates that this method, while simple to implement, suffers from limited adaptability, resulting in low efficiency in dynamic driving cycles, such as those encountered in urban conditions [11–13]. Conversely, torque-based algorithms activate the RBS when the motor generates negative torque. An example is the solution described in [14], which increased recovered energy by 5% in simulations, although it requires precise control to prevent wheel lock, as confirmed by analyses in [15].

A further strategy involves utilizing braking system pressure. Studies, such as that in [16], indicate that this method, when applied to an EV, led to a 1.22% decrease in battery consumption under NEDC testing, though its effectiveness is reduced when the SOC surpasses 80%. An enhanced approach involves the use of an ideal braking force distribution curve to modulate the regenerative and mechanical braking proportions in accordance with the vehicle’s current state and driving cycle. The findings of study [17] showed energy recovery levels reaching 79% for gentle braking (1–2 m/s^2) and 29% for medium-intensity braking (3–4 m/s^2), which was validated through simulations in [18]. The differences between these results imply a dependence on testing methodologies, thus emphasizing the requirement for greater standardization [19].

Crucial control parameters involve the brake pedal signal (indicating driver intent), SOC (which restricts energy absorption), and road conditions (such as slope and traction). Research [20] illuminates the significance of driver historical data and kinematic variables, such as longitudinal acceleration and slip ratio, in adaptive control strategy design. Yet, a research gap is evident, as the literature seldom examines the relative importance of these parameters in different operational contexts.

Mathematical models are employed by optimization algorithms to optimize energy recovery, all while ensuring driving safety and comfort. Among the prevalent approaches are fuzzy logic, genetic algorithms, and dynamic programming. For example, a strategy integrating fuzzy logic and genetic algorithms, as presented in [21] yielded a 10% enhancement in recovery efficiency and an 8% increase in EV range within urban driving cycles. In a contrasting approach, torque optimization, as outlined in [22], enhanced the recovery

coefficient by 3.35% in WLTC tests, minimizing energy wastage. However, study [23] conclude that the computational load of these methods limits their deployment in mass-market vehicles, notably within the budget-friendly segment.

Dynamic programming (DP) is also a technique that is used widely. In [24] DP achieved a 65% recovery rate during laboratory simulations, although this required simplifying the vehicle model, which reduces its practicality in real-world conditions. Neural networks, which were used in [25], improved RBS adaptation to changing road conditions, leading to a 5,3% increase in recovered energy compared to conventional PID controllers. The high computational demands of these approaches highlight the need to investigate simpler alternative solutions [26].

Predictive algorithms, particularly Model Predictive Control (MPC), employ real-time data, such as route topography from GPS and driving style, to forecast braking needs and optimize regenerative force. As demonstrated in [27] MPC resulted in a 15% increase in recovered energy in urban conditions through precise adaptation to velocity profiles.

The application of machine learning is becoming increasingly prevalent. Neural networks, as demonstrated in [28] effectively tailored the RBS strategy to individual driving patterns, yielding a 12% gain in efficiency during real-world trials. Type-2 fuzzy logic, when employed in [29], facilitated precise wheel slip management on low-traction surfaces, improving stability and energy recovery. Nevertheless, the sensitivity to input data quality (e.g., GPS) and the necessity for significant hardware resources limit the scalability of these approaches, as noted in [30,31].

Recent advancements highlight the integration of RBS with Vehicle-to-everything (V2X) communication. An example of energy consumption optimization in electric vehicles is presented, achieved through the targeted control of energy usage in critical areas. The methodology employs a neural network algorithm, integrating three communication protocols: vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2B), and vehicle-to-network (V2N). The final comparative analysis, which evaluated energy gains and other pertinent metrics, identified the V2B communication method as the most effective [32]. An energy-optimal braking strategy (EOBS) for electric vehicles is presented in [33], employing V2X communication to plan braking speed and maximize energy recapture. Simulation studies, performed using MATLAB and CarSim, revealed a substantial improvement in energy recovery compared to standard braking methodologies. The study [34] introduces the RT-EDPS system, which employs map and traffic light data to plan vehicle speed for optimized energy recovery. Upon detecting a preceding vehicle, the system initiates V2V communication to dynamically adjust the trajectory, demonstrating energy recovery gains exceeding 40% in simulation scenarios when compared to human drivers.

Based on the preceding analysis, it is evident that Regenerative Braking Systems have progressed significantly over time, transitioning from basic PID controllers to sophisticated artificial intelligence-driven systems. In dynamic urban settings, predictive algorithms have proven more effective than optimization algorithms, resulting in increased recovered energy. However, the variation in results underscores the absence of consistent testing standards, thereby complicating comparative analyses. Despite the potential benefits of adaptive algorithms, long-term effects, such as the deterioration of RBS components and the impact of driving style on efficiency, are rarely analyzed in the current literature. Also, research on integrating RBS with autonomous vehicles, where predictive systems could significantly enhance energy management, is lacking.

5.2. Safety and Stability Enhancements in Regenerative Braking Systems

5.2.1. Brake-by-Wire System

To address the safety and stability of vehicles with regenerative braking systems, the literature proposes several solutions and strategies. One of the foremost approaches is the implementation of brake-by-wire systems.

Present-day brake-by-wire systems constitute a sophisticated technological solution that replaces the conventional hydraulic link between the brake pedal and calipers with electronic governance. This design permits precise, independent allocation of braking force to each wheel, significantly boosting vehicle stability, particularly in critical instances such as abrupt braking maneuvers. In the context of electric vehicles, these systems are especially advantageous due to their integration of regenerative braking functionality. During deceleration, the electric motor operates as a generator, recovering kinetic energy and feeding it back into the battery, thus enhancing the vehicle's energy efficiency. Through electronic controls and advanced control algorithms, brake-by-wire systems dynamically manage the balance between mechanical and regenerative braking, optimizing energy recovery without compromising safety. The precise control of individual wheels minimizes the risk of traction loss, particularly on slippery surfaces, further enhancing the performance of ABS and Electronic Stability Control systems. Additionally, intelligent braking force management enables the system to adapt to varying driving styles and road conditions, contributing to a more comfortable and safer driving experience. Figure 6 illustrates the schematic of a brake-by-wire system in electric vehicles.

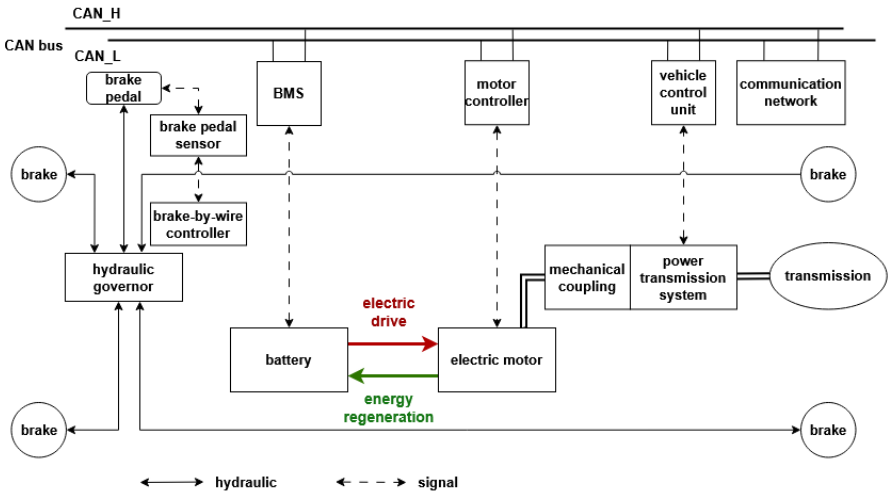


Figure 6. Brake-by-wire system in electric vehicle.

The ability of brake-by-wire systems to dynamically and precisely distribute braking force between regenerative electric braking and hydraulic friction braking contributes to increased energy recovery efficiency and enhanced vehicle stability. A substantial body of literature is dedicated to the development of control strategies for managing braking force distribution between the hydraulic system and the electric motor [35–37]. The coordination of regenerative and hydraulic braking is also extensively addressed in various studies, which utilize hierarchical control strategies to balance energy recovery maximization with vehicle stability [38,39]. Dynamic adjustment of braking force to varying road conditions and improved stability are achieved through the use of these systems within anti-lock braking systems [40]. The integration with energy recovery functions is a focal point in the literature, as it boosts vehicle energy efficiency and enhances both safety and driving comfort. Nevertheless, studies indicate technical concerns, including the dependence on electronic component reliability, which requires ongoing research and development.

5.2.2. Regenerative Braking with In-Wheel Motors

Numerous studies introduce the concept of electric drive systems utilizing in-wheel motors, where electric machines are integrated directly into the wheel hubs. During braking, these in-wheel electric machines generate electromagnetic resistance, converting the vehicle's kinetic energy into electrical energy, which is then stored in the battery. Due to the variable distribution of braking force between axles, depending on load transfer, electric vehicles with in-wheel motors exhibit a higher energy recovery rate during braking than two-axle drive vehicles. The findings of study [41] indicate that all-wheel drive electric vehicles facilitate energy recovery during braking that is 23% and 31% greater, respectively, when compared to front-wheel or rear-wheel drive electric vehicles.

Two significant factors are paramount in improving the energy recovery rate during braking for electric vehicles with in-wheel motors: the distribution of braking force between the front and rear axles, and the allocation of braking force between the electric motor and mechanical brakes on each axle. A critical component in enhancing energy recovery for this type of electric vehicle is maximizing the participation and efficiency of electric motor-driven braking in generator mode. To achieve optimal kinetic energy recovery, a coordinated distribution of braking force, reflective of the vehicle's braking dynamics and the complex braking system's attributes, is required [42,43].

Effective control of in-wheel electric motors in electric vehicles is essential for maximizing energy recuperation during braking. The literature offers a range of strategies for optimizing this procedure. Advanced control systems, such as algorithms employing adaptive control theory and genetic algorithms, allow for the dynamic modulation of regenerative and mechanical torque, optimizing energy recovery and ensuring user safety. As an example, study [44] presented a control strategy for a brushless direct current motor equipped with multi-stage bidirectional converters. This strategy relied on the accurate estimation of drive shaft torque through the evaluation of forces acting on an electric vehicle traversing an inclined road. The findings indicated that implementing this strategy effectively manages power flow and reduces harmonic distortion, both of which are critical for maximizing energy recovery. Study [45] detailed the design of a control system that incorporates motor and battery temperature, along with their state of charge, to enable dynamic energy recovery adjustment and protect components from thermal overload. A motor speed optimization system, based on a particle swarm optimization (PSO) algorithm integrated with a PID controller, was proposed in study [46] leading to significant improvements in energy recovery efficiency and battery longevity. Study [47] presented a two-stage nonlinear predictive control algorithm, which dynamically distributes braking force between regenerative and mechanical braking systems, resulting in a range increase of over 24% for electric vehicles, while maintaining safety and optimizing battery life.

The significance of integrating advanced control algorithms and adaptive optimization methods to optimize energy recovery while upholding the safety and comfort of electric vehicles with in-wheel motors is consistently emphasized in published studies. Researchers are increasingly focusing on the critical role of predictive models, which are based on machine learning, in real-time optimization of braking system operation, and these models consider dynamically changing road conditions and driving behavior [48–50]. Predictive models can also enhance autonomous driving systems by providing more complex vehicle trajectory analysis and road situation forecasting. This opens the way for the development of control systems that can operate proactively, reducing energy losses and the risks associated with abrupt changes in road conditions.

5.2.3. Integration of ABS with Regenerative Braking: Balancing Safety and Energy Efficiency

In vehicles utilizing regenerative braking energy recovery systems, the creation and deployment of advanced braking torque distribution strategies are paramount for achieving optimal energy efficiency and vehicle safety. These strategies, as mentioned earlier, must address dynamically changing conditions, including diverse road adhesion coefficients (e.g., dry, wet, or icy surfaces), which impact the wheels' ability to generate braking forces without causing slippage. A multitude of publications detail authors' strategies for distributing regenerative and mechanical braking force to regulate wheel slip ratios. The control strategies found in the literature can be generally classified into two main types: optimization algorithm-based and real-time adaptive [48,49].

Optimization strategies rely on offline methodologies to compute optimal reference values and trajectories, with the objective of maximizing system performance. In this context, dynamic programming and particle swarm optimization (PSO) are frequently utilized. By analyzing historical data, optimization strategies allow for fine-tuned brake control adjustments to suit specific driving conditions, thereby optimizing braking efficiency. Study [50] provides an example with the "Hybrid-ABS" concept, in which optimization algorithms are used to divide the braking torque between mechanical brakes and the regenerative system. This approach not only reduces braking distance, but also reduces the stress on the traditional braking systems, which increases driving comfort and extends the lifespan of the mechanical components. Study [51] introduced a strategy that combines particle swarm optimization and ant colony optimization. The simulations showed that the suggested strategy not only ensures safety in emergency situations, but also allows for a significant improvement in braking energy recovery. Energy recovery in the urban cycle increased by 16% compared to traditional methods. A disadvantage to optimization strategies is that the applied control algorithms have high computational requirements, which limits their adaptation in various braking conditions.

Real-time adaptive strategies achieve coordinated control by applying diverse algorithms, including proportional-integral-derivative (PID) control algorithms, linear-quadratic regulator (LQR) algorithms, and strategies based on model predictive control (MPC) [52].

The proportional-integral-derivative (PID) controller, a cornerstone of automation, is extensively applied for dynamic process regulation. It employs three components—proportional (P), integral (I), and derivative (D)—that collectively strive to minimize the discrepancy between the setpoint and the actual value. To achieve coordinated control of regenerative and ABS systems in a PID control strategy, the PID controller's parameters are adjusted according to the vehicle's current kinematic parameters and designated control objectives. This strategy is distinguished by its operational simplicity and low computational demands. Despite these advantages, it yields only suboptimal control laws, which complicates the precise and efficient management of the relationship between the regenerative braking system and the ABS system. An example of the application of this strategy is shown in study [53]. It shows a PID controller model for an ABS in an electric vehicle that significantly reduces braking distance and controls wheel slip.

Widely implemented in dynamic systems, the Linear-Quadratic Regulator (LQR) is an advanced optimal control algorithm that seeks to minimize a defined cost function while stabilizing the system. LQR effectively balances system performance efficiency against control cost. In electric vehicles, LQR facilitates optimized energy recovery during regenerative braking, concurrently minimizing control costs However, the LQR method necessitates a linearization process to simplify the control challenge and lacks the capability to adjust weights assigned to distinct time steps within the prediction horizon, potentially limiting the impact of coordinated braking force distribution.

Model Predictive Control (MPC) is an advanced control strategy used in dynamic systems, including regenerative braking and ABS management for electric vehicles. By using a mathematical model to predict the system's future behavior, MPC is able to make dynamic control decisions. The system's response to various control signals is forecast by the MPC mathematical model. The Strategy works in real time, by predicting future system states over a fixed time horizon at each step, and then adjusting the control, and moving forward step by step. As an example, researchers in study [54] devised an adaptive control model that dynamically adjusts braking force distribution between the regenerative and mechanical braking systems in real-time, considering the instantaneous wheel slip ratio. This enables the achievement of optimal energy recovery efficiency before the engagement of the ABS system, while preserving vehicle stability. Study [55] developed a simulation model for braking force distribution, which accounts for the dynamic properties of the vehicle and the variability of road conditions, thereby improving both energy recovery efficiency and vehicle stability across various scenarios. In study [56] an algorithm was proposed to estimate parameters such as master brake cylinder pressure and valve block flow, aiming to improve the smoothness of coordinated control. Moreover, study [57] demonstrated integrated control of the anti-lock braking system and the regenerative system, showing a 52.9% increase in recovered energy during ABS braking conditions through the use of adaptive algorithms.

Although Model Predictive Control offers a range of advantages, it also presents several limitations. Firstly, it demands the creation of a highly accurate model of the system being controlled. Errors in this model can lead to control inaccuracies [58,59]. Secondly, the method's reliance on significant computational resources can be problematic for systems with limited processing power. Lastly, in highly dynamic applications, the time needed to solve the optimization problem may be insufficient [60–62].

To summarize, the evolution of advanced braking torque distribution strategies in vehicles utilizing regenerative braking energy recovery systems is a critical progression towards maximizing energy efficiency and ensuring driving safety. These strategies are generally divided into optimization-based techniques, which provide high precision and effectiveness but demand considerable computational power, and adaptive techniques, which emphasize real-time responsiveness and flexible adjustment to varying road conditions. Notwithstanding the benefits of each method, their practical application requires continued research to address challenges related to computational load and model precision. The synergistic combination of these strategies within integrated systems can lead to a marked improvement in energy efficiency, driving comfort, and safety.

5.3. Energy Storage Technologies for Regenerative Braking Applications

Extensive literature documents examples of electric drive systems supplemented with auxiliary energy storage units dedicated to the storage of braking-recovered energy (Figure 7). These systems are referred to as Hybrid Energy Storage Systems (HESS). They are implemented to attain an optimal combination of energy density and power capability [63–65]. The power capacity of the battery significantly influences the energy recovery dynamics during braking. The specifications of the battery integrated into the system define and restrict the amount of electrical energy it can accommodate within a short duration. The integration of two batteries with distinct specifications enables the provision of both high power and energy capacity, thereby enhancing the efficiency of the regenerative braking system.

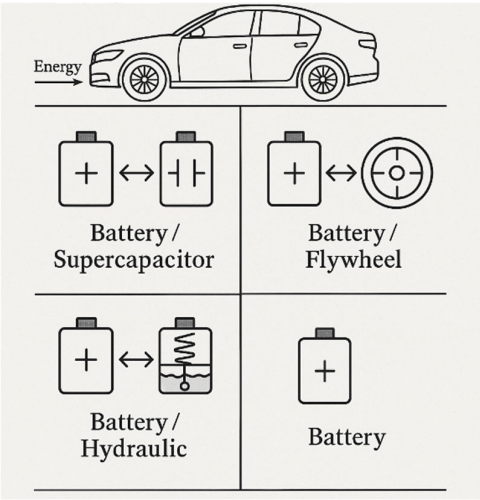


Figure 7. HESS in regenerative braking of an EV.

The literature presents, as one example, a system that utilizes an electrochemical battery as the primary energy source, and a supercapacitor as a supplementary energy storage device. Lithium-ion batteries provide the capacity for large energy storage, while supercapacitors allow for rapid energy acceptance. The stored energy is then used to power the vehicle. Supercapacitors are particularly useful, due to their high number of operational cycles, which can reach millions, for managing short-term power peaks, such as during intense braking or dynamic acceleration [66–68]. However, because of their limited energy density, they are more appropriate for short term energy storage, which makes them an ideal complement to lithium-ion batteries.

Beyond electrochemical batteries, other energy storage solutions include kinetic energy storage devices (e.g., Kinetic Energy Recovery Systems, KERS) and hydropneumatics accumulators. Kinetic accumulators utilize rapidly rotating masses to store kinetic energy. Their high durability and efficiency make them ideal for short-duration energy storage and release. Despite their high overload resistance and quick response times, the higher implementation costs and vibration sensitivity of kinetic accumulators may limit their application in mass production. The analysis in study [69] indicates that a hybrid system combining an ion battery with a kinetic accumulator can significantly improve the energy recovered during braking.

The principle of hydropneumatics accumulators involves using gas pressure to store energy through fluid compression. During braking, recovered energy is stored as the elastic energy of the gas, which is then released via expansion. This method is beneficial for its high tolerance to variable loads and extensive cycle lifespan. However, the system's large size and weight are notable drawbacks. Study [70] shows that implementing a hydraulic regenerative braking system in an all-electric vehicle can significantly boost acceleration and increase driving range by approximately 28%. Furthermore, simulations suggest that a regenerative system with a hydropneumatics accumulator, combined with an appropriate energy management strategy, can achieve energy recovery improvements of up to 16.73% at various velocity.

Although energy recovery systems with additional energy storage demonstrate significant potential, their widespread integration in present-day electric vehicle models has yet to be realized. The primary constraints include the technological complexity involved, the substantial mass of these systems, and the necessity for advanced control strategies that rely on sophisticated algorithms to coordinate the operation of diverse energy storage components [71–73]. While the combination of supercapacitors or flywheels with lithium-ion

batteries in HESS systems offers considerable benefits, the high production costs associated with these solutions restrict their widespread implementation in mass-produced vehicles.

Hybrid energy storage systems play a vital role in the efficient utilization of energy recuperated during regenerative braking in electric vehicles. By integrating a range of energy storage technologies, HESS delivers flexibility, efficiency, and durability, while also minimizing energy dissipation and maximizing the lifespan of critical system components. Dynamic adaptation of energy management strategies to changing driving conditions and personalized vehicle needs is enabled through the optimization of the collaborative function of different energy storage devices within HESS.

The prospective development of HESS is expected to contribute to further improvements in vehicle energy efficiency, concurrently supporting sustainable development and increased market penetration. However, technical limitations, substantial implementation costs, and the absence of standardized design specifications pose considerable obstacles to their broader adoption. It is posited that the development of more affordable components and improved computational algorithms will facilitate the proliferation of HESS in commercial electric vehicle models.

5.4. Impact of Environmental, Route, and Kinematic Factors on Energy Recovery

The efficiency of regenerative braking systems within electric vehicles is significantly dependent upon a variety of variables, including environmental conditions, route parameters, and vehicle kinematic parameters (Figure 8). Environmental conditions impact the operational efficiency of drivetrain components and the vehicle's dynamic behavior, which directly influences regenerative efficiency. Low temperatures, notably, reduce the efficiency of lithium-ion batteries, thus diminishing their capacity to store energy recovered during braking. Studies have shown that both low and high ambient temperatures negatively affect energy recovery and its overall performance. Electric vehicle tests on a chassis dynamometer across various temperatures, conducted by the authors of study [74] showed significantly reduced energy recovery efficiency at sub-zero temperatures. Consistent results were found in study [75], which, through real-world testing of three electric vehicles in Central European climatic conditions, revealed that regenerative braking efficiency is strongly dependent on battery temperature, particularly below 10°C, and may be fully disabled at −4°C.

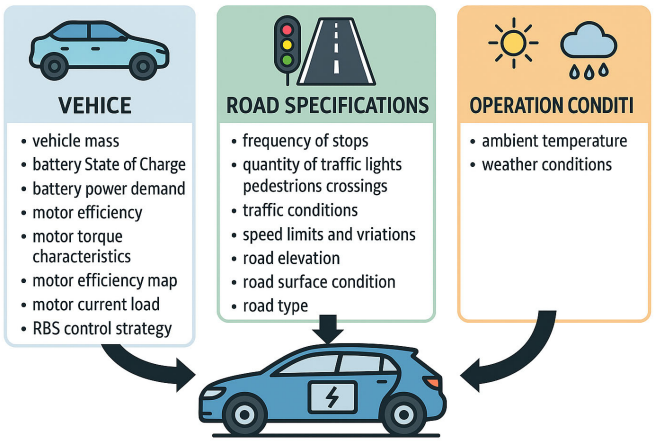


Figure 8. Overview of factors influencing regenerative braking energy recovery in electric vehicles.

Atmospheric precipitation, such as rain, snow, or ice, significantly alters vehicle traction parameters. This results in increased rolling resistance and a diminished adhesion coefficient, necessitating a greater reliance on mechanical braking systems and consequently limiting regenerative braking efficiency. Research data demonstrates a 10–20% decrease in

energy recovery when operating on wet road surfaces, as compared to ideal dry surface conditions [76]. This is caused by a diminished capacity to efficiently convert the vehicle's kinetic energy into electrical energy, a key process in recuperation. Furthermore, fog and intense wind conditions add additional variables to driving dynamics, which may have an indirect impact on recuperation. However, the influence of these factors on energy recovery efficiency requires further research and more in-depth analysis. Atmospheric conditions are a critical factor affecting recuperation efficiency in electric vehicles. Consequently, understanding and modeling these dependencies is essential for optimizing recuperation systems and improving vehicle energy efficiency in a variety of operational contexts.

The topographical characteristics of a route, along with its surface properties, are primary factors affecting braking frequency and the amount of kinetic energy that can be recuperated. Study [77] indicates that the efficiency of regenerative braking systems in electric vehicles is strongly correlated with the ratio of electric motor power and battery capacity to vehicle mass, and significantly influenced by terrain conditions, such as changes in altitude. Simulation results, as presented in study [78] confirm that terrain slope has a substantial impact on the efficiency of the recuperation process. Energy recovery is increased during downhill driving, while energy consumption increases during uphill driving. Real-world traffic studies, as in study [79] how that road topography, initial braking velocity, and the duration of brake pedal application are the main factors affecting recovered and lost energy. Literature data indicates that routes with a 5–10% slope result in 40–60% higher energy recovery compared to flat terrain [80]. This is due to the efficient harnessing of gravitational forces for electrical energy generation. However, recuperation becomes nearly impossible on uphill sections, increasing energy demand and thereby reducing the average energy recovery efficiency over the complete driving cycle.

The road surface is also an important factor in the recuperation process. Surfaces that have a low rolling resistance coefficient, for example asphalt, reduce energy losses, and therefore increase the proportion of kinetic energy available for recovery. On the other hand, uneven surfaces, which include unpaved roads, generate higher energy losses due to rolling resistance, and this results in a reduction of 5–15% in recuperation efficiency [81].

The efficacy of regenerative braking systems in electric vehicles is strongly correlated with operational conditions. In urban environments, which are characterized by dynamic driving and frequent acceleration and braking cycles, the kinetic energy recovery system can achieve high efficiency. It is estimated that, under these circumstances, recovered braking energy can represent 20 to 40% of the energy consumed for vehicle propulsion [82]. Comparable conclusions are found in study [83]. Real-world electric vehicle tests have shown that the level of recovered energy changes depending on driving conditions, specifically driving velocity and traffic intensity. Based on the analyses presented in [84] the authors conclude that a lower average velocity is associated with a higher proportion of unit regenerative braking energy, which results from typical urban traffic conditions, such as frequent stops and braking phases.

Differing from urban driving conditions, vehicle operation on highway routes, where constant velocity driving is predominant, results in minimal energy recuperation. In such scenarios, due to the infrequent incidence of braking maneuvers, recuperation efficiency is substantially lower, often not exceeding 5% of the energy consumed for propulsion [82]. The low recuperation efficiency during highway driving is attributed to the limited frequency of braking. Additionally, during instances of abrupt deceleration, the recuperation system may restrict its operation due to an excess of kinetic energy, which prevents full energy recovery.

Vehicle motion parameters, specifically velocity, acceleration, and braking intensity, dictate the magnitude of generated and potentially recoverable kinetic energy. As kinetic energy is proportional to the square of velocity, recuperation potential escalates with

increasing vehicle velocity. Nevertheless, recuperation systems are subject to constraints. In scenarios involving rapid deceleration from high velocity (e.g., above 80 km/h), surplus kinetic energy may be dissipated as thermal energy, leading to decreased energy recovery efficiency. Study [85] suggests that the most effective initial braking velocity for energy recuperation is between 20–50 km/h.

The findings of study [86], suggest that maximizing energy recovery is achieved through smooth braking with a deceleration of around 1–2 m/s^2. In contrast, abrupt braking, defined by deceleration rates above 3 m/s^2, often triggers mechanical braking systems, resulting in reduced recuperation efficiency. Publications [87,88] indicate that braking intensity significantly influences energy recovery. Elevated braking intensity correlates with an increase in regenerative braking force, which facilitates the system's ability to utilize a larger amount of regenerative braking energy.

Scientific studies demonstrate that employing adaptive braking strategies can lead to a 15–25% improvement in recovered energy [89]. By optimizing braking parameters according to driving conditions, these strategies enable the recuperation system to be used more efficiently.

Regenerative braking systems in electric vehicles exhibit energy recovery levels that are collectively determined by operating conditions, route characteristics, and kinematic parameters. The highest efficiency is generally observed on routes with variable slopes, under moderate weather conditions, and during smooth driving with frequent, gentle braking maneuvers. However, challenges such as extreme temperatures, uneven terrain, and highway driving limit the potential for energy recovery. Future research should prioritize integrating environmental and kinematic data to develop more advanced regenerative systems that can adapt to varying operational conditions.

Advancements in braking energy recuperation systems include the development of predictive systems that utilize data from GPS, road maps, and artificial intelligence algorithms to optimize energy recovery in real-time. The integration of these technological solutions allows for increased recuperation efficiency through the adaptation of braking strategies to forecasted route and environmental conditions. By analyzing data pertaining to route profiles, traffic volume, and atmospheric conditions, these systems can predict optimal moments for regenerative braking, leading to an enhancement in recovered energy.

6. Discussion

The undertaken literature review on energy recovery systems in electric vehicles illustrates significant development in this sector. This advancement is spurred by the increasing necessity for heightened energy efficiency and the promotion of sustainable transport solutions. Key findings demonstrate that the efficiency of recuperation is substantially reliant on the interaction between environmental influences, route properties, and kinematic parameters.

A critical review of the current literature highlights several limitations in existing research. First, while control strategies like predictive and adaptive algorithms are capable of increasing recovered energy, their computational complexity and significant hardware demands restrict their application in mass-produced vehicles, especially those intended for budget-conscious consumers. A comparison of parallel and series strategies reveals that series configurations offer improved flexibility for ABS integration, but at the expense of increased technical complexity, raising the issue of the optimal balance between efficiency and cost-effectiveness.

Technologies for energy storage, including hybrid HESS systems, have shown promise in improving energy recovery by integrating supercapacitors and lithium-ion batteries. Nevertheless, their restricted adoption, primarily due to high costs and control complexities, indicates that more research is needed to determine the economic scalability of these solutions.

Furthermore, the literature largely centers on electric vehicles and hybrid electric vehicles, while other vehicle types, such as autonomous vehicles and fuel cell electric vehicles, receive insufficient attention, despite their increasing relevance in the automotive industry.

Ensuring the integration of recuperation systems with requisite safety standards remains a significant challenge. Brake-by-wire systems and in-wheel motors, while offering precise braking control and thus improving stability and energy recovery efficiency, necessitate further research to ascertain their reliability in extreme environmental conditions. Additionally, the effect of low temperatures on battery performance, evidenced by reduced recovered energy values in cold ambient temperatures, indicates a need for the development of materials with increased thermal tolerance.

The review identifies several research gaps that warrant further investigation. Firstly, there is a notable absence of comparative studies assessing the long-term degradation of RBS components across various operational conditions. Secondly, insufficient attention is given to autonomous vehicles, where predictive control strategies could significantly enhance energy recovery. Thirdly, the influence of human factors, particularly driving style, on recuperation efficiency is not adequately understood, despite its potential implications for the development of adaptive algorithms.

The practical implications of these findings are particularly relevant for electric vehicle system designers. To optimize recuperation effectively, a comprehensive approach is required, integrating GPS data, artificial intelligence, and advanced materials to enable systems to adapt to fluctuating environmental and route conditions. Concurrently, reducing the cost of technologies such as HESS and brake-by-wire can significantly expedite their commercialization, thereby enhancing the range and efficiency of the electric drive system. Researchers are advised by these results to undertake interdisciplinary studies that combine mechanical engineering, computer science, and materials engineering to address the complex technological and environmental challenges.

7. Conclusions

A comprehensive literature review on energy recovery systems in electric vehicles confirms their fundamental role in enhancing energy efficiency and range. The analysis of publications from 2005 to 2024 illustrates that recuperation efficiency is dependent on a multitude of factors, including environmental conditions, route topography, and kinematic parameters. Technological advancements, encompassing the implementation of predictive algorithms and hybrid energy storage systems (HESS), display potential for improving energy recovery efficiency. The integration of brake-by-wire systems and in-wheel motors enhances vehicle stability and energy recovery, which is particularly critical for safety in varied road conditions. However, their widespread adoption is restricted by high costs and implementation complexity. Moreover, the literature highlights research gaps, such as the inadequate analysis of the effects of environmental factors and operating conditions, the long-term degradation of Regenerative Braking System (RBS) components, and the application of recuperation in autonomous and fuel cell electric vehicles.

The conclusions drawn from this review underscore the necessity for continued technological and research innovation to fully exploit the potential of recuperation in modern transport. The following recommendations are presented: (1) comprehensive studies examining the influence of atmospheric conditions on energy recovery; (2) the development of cost-effective Hybrid Energy Storage Systems and brake-by-wire technologies; (3) the investigation of predictive control strategies for autonomous vehicles. Executing these recommendations can significantly enhance electric vehicle efficiency, thereby supporting their competitiveness and the sustainable evolution of the automotive sector.

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